

Higher projective maximal orthogonal families of measures

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Abstract

Let $1 \leq n < \omega$ and assume that the canonical mouse M_n with n Woodin cardinals exists. We describe a forcing extension of M_n in which every boldface Σ_{n+2}^1 set of reals has the Baire property, boldface Σ_{n+2}^1 -uniformization holds, and there is a lightface Π_{n+2}^1 maximal orthogonal family of Borel probability measures on 2^ω . Additionally there is no boldface Σ_{n+2}^1 maximal orthogonal family in the same model.

1 Introduction

Let $P(2^\omega)$ be the space of Borel probability measures on Cantor space. Two measures μ and ν are orthogonal, written $\mu \perp \nu$, if they are concentrated on disjoint Borel sets. A maximal orthogonal family, abbreviated *mof*, is a maximal set of pairwise orthogonal measures.

Fischer and Törnquist proved that $V = L$ implies the existence of a Π_1^1 mof [FT10]. Fischer, Friedman, and Törnquist later obtained models with a Π_2^1 mof and no Σ_2^1 mof [FFT12]. The present argument applies the same measure coding method at the level determined by M_n .

The forcing uses three kinds of stages. At a mof stage, a real coding a measure is considered. If the measure is orthogonal to all measures accepted earlier, a code is written into a fresh block of Suslin trees. The coding stage uses the method of Fischer, Friedman, and Törnquist: the almost disjoint generic u selects the columns in $\Delta(u)$, and the same real u is read from the measure code [FFT12, FFZ11]. At Cohen stages, a Cohen real is added. At category amoeba stages, a Borel meager set is added which covers all Borel meager sets coded in a prescribed inner model. The latter two kinds of stages give the Baire property for all boldface Σ_{n+2}^1 sets.

To prove the Σ_{n+2}^1 -uniformization theorem, let N be a small generic extension of M_n and let $s \in \mathbb{R}^N$. The canonical mouse $M_n(s)$ carries a good $\Sigma_{n+2}^1(s)$ well-order of its reals. Steel capture puts witnesses for $\Sigma_{n+2}^1(s)$ statements into $M_n(s)$. Choosing the least witness in the mouse order gives a uniformizing relation of the same complexity.

We prove the following theorem.

Theorem 1.1. *Let $1 \leq n < \omega$ and assume that M_n exists. There is a cardinal-preserving forcing extension V^* of M_n such that:*

- (1) $V^* \models \text{CH}$;

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- (2) every boldface Σ_{n+2}^1 set of reals has the Baire property;
- (3) boldface Σ_{n+2}^1 -uniformization holds;
- (4) there is a lightface Π_{n+2}^1 maximal orthogonal family of Borel probability measures on 2^ω ;
- (5) there is no boldface Σ_{n+2}^1 maximal orthogonal family of Borel probability measures on 2^ω .

All pointclasses in the regularity and uniformization statements are boldface. The mof constructed in the proof is lightface definable. We use standard recursive pairings of reals without further comment.

2 Measures and orthogonality

We recall the facts about measures used below. See [FT10, FFT12] for details.

Let

$$p(2^\omega) = \{f : 2^{<\omega} \longrightarrow [0, 1] : f(\emptyset) = 1 \text{ and } f(s) = f(s \smallfrown 0) + f(s \smallfrown 1)\}.$$

For $f \in p(2^\omega)$, let μ_f be the measure determined by

$$\mu_f(N_s) = f(s),$$

where $N_s = \{x \in 2^\omega : s \subseteq x\}$. This gives a recursive presentation of $P(2^\omega)$. Let

$$p_c(2^\omega) = \{f \in p(2^\omega) : \mu_f \text{ is non-atomic}\}.$$

The set $p_c(2^\omega)$ is arithmetical. The relations

$$\mu_f \perp \mu_g, \quad \mu_f \ll \mu_g, \quad \mu_f \approx \mu_g$$

are arithmetical on measure codes.

Lemma 2.1 (Fischer–Törnquist coding lemma). *There are a recursive partial function*

$$r_{\text{FT}} : \text{dom}(r_{\text{FT}}) \longrightarrow 2^\omega$$

and a recursive function

$$G : p_c(2^\omega) \times 2^\omega \longrightarrow p_c(2^\omega)$$

such that, for every $f \in p_c(2^\omega)$ and $u \in 2^\omega$,

$$\mu_{G(f,u)} \approx \mu_f \quad \text{and} \quad r_{\text{FT}}(G(f,u)) = u.$$

The function changes conditional weights along a recursive sequence of splitting nodes. The bits of u are read from the resulting ratios. Absolute equivalence with the original measure is preserved.

We also use the Kechris–Sofronidis map. Define

$$xE_Iy \iff \sum_{k < \omega} \frac{|x(k) - y(k)|}{k+1} < \infty.$$

Fact 2.2 (Kechris–Sofronidis). *There is a Borel map $x \mapsto \mu^x$ from 2^ω to $P(2^\omega)$ such that*

$$xE_Iy \implies \mu^x \approx \mu^y, \quad \neg(xE_Iy) \implies \mu^x \perp \mu^y.$$

The equivalence relation E_I is turbulent and each E_I -class is meager [KS01, Hjo00].

For a fixed measure μ , the relation $\ll$ has the ccc below μ : every family of pairwise orthogonal nonzero measures which are absolutely continuous with respect to μ is countable. We use this in the proof that no Σ_{n+2}^1 mof exists.

3 The input from M_n

Fix $1 \leq n < \omega$. Let δ_0 be the least Woodin cardinal of M_n . We use the following facts about M_n . They are consequences of the projective analysis of the canonical mice and of the coding construction over M_n ; see [Ste95, Ste10, Hof26].

Fact 3.1. *The following objects can be fixed in M_n .*

- (1) *A canonical sequence*

$$\vec{S} = \langle S_\xi : \xi < \omega_1 \rangle$$

of independent Suslin trees. Thus, for every finite $e \subseteq \omega_1$, the product $\prod_{\xi \in e} S_\xi$ is a Suslin tree.

- (2) *A canonical matrix*

$$\vec{d} = \langle d_{\xi, m} : \xi < \omega_1, m < \omega \rangle$$

of pairwise almost disjoint infinite subsets of ω .

- (3) *A projective class of codes for the correct countable initial segments of the canonical M_n -construction. The assertion that a real codes such an initial segment is Π_{n+1}^1 .*

- (4) *A canonical Martin–Solovay tree T_{n+1} representing the universal Σ_{n+2}^1 set. Its representation, together with its complementing tree, is absolute to the forcing extensions of size below δ_0 used below.*

Moreover, $M_n \models \text{CH}$.

We do not reproduce the fine structural definition of the correct initial segments. We use only their projective complexity, comparison, and condensation. A countable transitive model containing such an initial segment computes the corresponding initial part of $\vec{S}$ and $\vec{d}$ correctly.

4 Coding with fresh Suslin-tree blocks

This section defines the coding operation used at mof stages. The use of $\Delta(u)$ follows the coding method in [FFT12, FFZ11].

4.1 Columns in one fresh block

For $\eta < \omega_1$, let

$$\vec{S}^\eta = \langle S_{\omega\eta+j} : j < \omega \rangle.$$

Fix a recursive bijection

$$\rho : (\omega \setminus \{0\}) \times \omega \times 2 \longrightarrow \omega.$$

For $m > 0$, $k < \omega$, and $i < 2$, put

$$S_{k,i}^{\eta,m} = S_{\omega\eta+\rho(m,k,i)}.$$

Thus the block $\vec{S}^\eta$ is split into columns indexed by the positive integers. Each column is split into pairs

$$\{S_{k,0}^{\eta,m}, S_{k,1}^{\eta,m}\} \quad (k < \omega).$$

Let $z \in 2^\omega$. For $m > 0$, define

$$\mathbb{B}_{\eta,m}(z) = \prod_{k < \omega}^{\text{fin}} S_{k,z(k)}^{\eta,m}.$$

If $H_m \subseteq \mathbb{B}_{\eta,m}(z)$ is generic, let

$$b_{m,k} = \bigcup \{p(k) : p \in H_m \text{ and } k \in \text{dom}(p)\}.$$

Then $b_{m,k}$ is a cofinal branch through $S_{k,z(k)}^{\eta,m}$.

Let

$$\mathbb{B}_\eta^*(z) = \prod_{m > 0}^{\text{fin}} \mathbb{B}_{\eta,m}(z).$$

Equivalently, this is the finite-support product of the trees

$$S_{k,z(k)}^{\eta,m} \quad (m > 0, k < \omega).$$

Lemma 4.1. *The forcing $\mathbb{B}_\eta^*(z)$ is ccc. It preserves every tree in $\vec{S}$ which is not one of the selected trees $S_{k,z(k)}^{\eta,m}$.*

Proof. There are countably many finite supports. On a fixed support, the forcing is a finite product of distinct members of $\vec{S}$. Such a product is ccc by independence. The same argument applied after adding one further tree gives the preservation statement. $\square$

4.2 The David code of one column

Fix $m > 0$. Let

$$\mathcal{B}_{\eta,m,z} = \langle b_{m,k} : k < \omega \rangle$$

be the branch sequence added by $\mathbb{B}_{\eta,m}(z)$. We code this sequence by a subset of ω_1 .

Let

$$m_\infty = M_n \upharpoonright \omega_1$$

be the ω_1 -initial part of M_n from which $\vec{S}$ and $\vec{d}$ are computed. In the branch extension, let $\lambda_{\eta,m,z}$ be least such that

$$N_{\eta,m,z} = L_{\lambda_{\eta,m,z}}[m_\infty, \eta, m, \mathcal{B}_{\eta,m,z}]$$

has size $\aleph_1$ and satisfies:

- (1) $N_{\eta,m,z} \models \text{ZFC}^- + \text{“}\aleph_1 \text{ exists”}$ and $\omega_1^{N_{\eta,m,z}} = \omega_1$;
- (2) $N_{\eta,m,z}$ computes $\vec{S}$ and $\vec{d}$ from m_∞ ;
- (3) $N_{\eta,m,z}$ verifies that the displayed branches record z in column m of block η .

Let $X_{\eta,m,z} \subseteq \omega_1$ be the fixed code for

$$(N_{\eta,m,z}, \in, m_\infty, \eta, m, \mathcal{B}_{\eta,m,z}).$$

Apply David's reshaping method to $X_{\eta,m,z}$. Choose the least λ such that

$$L_\lambda[X_{\eta,m,z}] \models \text{ZFC}^- + \text{“}\aleph_1 \text{ exists”}.$$

Inside this model, take the least continuous increasing sequence

$$\langle N_\xi : \xi < \omega_1 \rangle$$

of countable elementary submodels whose union is $L_\lambda[X_{\eta,m,z}]$. Put

$$c_\xi = N_\xi \cap \omega_1, \quad C = \{c_\xi : \xi < \omega_1\}.$$

The set $Y_{\eta,m,z} \subseteq \omega_1$ has two parts. Its odd part codes $X_{\eta,m,z}$. Its even part codes the increasing enumeration of C by interval well-orders.

Lemma 4.2 (David localization). *Let M be a countable transitive model of*

$$\text{ZFC}^- + \text{“}\aleph_1 \text{ exists”}$$

and suppose that $Y_{\eta,m,z} \cap \omega_1^M \in M$. Then M recovers the corresponding initial part of $X_{\eta,m,z}$. It also recovers a local transitive model containing:

- (1) *the correct M_n -initial segment at ω_1^M ;*
- (2) *column m of block η ;*
- (3) *the restrictions of the selected branches;*
- (4) *the statement that these branches record z .*

For later use, let $\dot{Y}_{\eta,m,z}$ denote the $\mathbb{B}_{\eta,m}(z)$ -name for this set.

4.3 The self-indexed coding forcing

For $u \in 2^\omega$, define

$$\Delta(u) = \{2k+2 : k \in u\} \cup \{2k+1 : k \notin u\}.$$

Thus $\Delta(u)$ contains exactly one member of each pair

$$\{2k+1, 2k+2\}.$$

If $s \in 2^{<\omega}$, put

$$\Delta(s) = \{2k+2 : k < |s| \text{ and } s(k) = 1\} \cup \{2k+1 : k < |s| \text{ and } s(k) = 0\}.$$

A condition in $\text{Code}_n(z, \eta)$ is a triple

$$p = (s_p, \pi_p, F_p)$$

with the following properties:

- (1) $s_p \in 2^{<\omega}$;
- (2) π_p is a condition in $\mathbb{B}_\eta^*(z)$ and

$$\text{supp}(\pi_p) \subseteq \Delta(s_p) \times \omega;$$

- (3) F_p is a finite subset of $\omega_1 \times \Delta(s_p)$;

(4) if $(\xi, m) \in F_p$, then

$$(\pi_p)_m \Vdash_{\mathbb{B}_{\eta,m}(z)} \xi \in \dot{Y}_{\eta,m,z}.$$

Here $(\pi_p)_m$ is the restriction of π_p to column m .

We put $q \leq p$ if:

- (1) $s_p \subseteq s_q$, $\pi_q \leq \pi_p$, and $F_p \subseteq F_q$;
- (2) if $(\xi, m) \in F_p$, then

$$\{j : |s_p| \leq j < |s_q| \text{ and } s_q(j) = 1\} \cap d_{\xi,m} = \emptyset.$$

The fourth clause in the definition of a condition is required for every member of F_q .

Let $K \subseteq \text{Code}_n(z, \eta)$ be generic. Put

$$u = \bigcup \{s_p : p \in K\}.$$

For $m \in \Delta(u)$, let H_m be the filter on $\mathbb{B}_{\eta,m}(z)$ generated by the column conditions $(\pi_p)_m$, $p \in K$.

Lemma 4.3 (Properties of the coding forcing). *The forcing $\text{Code}_n(z, \eta)$ is ccc and has size $\aleph_1$. In its generic extension:*

- (1) if $m \in \Delta(u)$, then H_m is $\mathbb{B}_{\eta,m}(z)$ -generic;
- (2) if $Y_{\eta,m,z}$ is the David code obtained from H_m , then, for every $\xi < \omega_1$,

$$\xi \in Y_{\eta,m,z} \iff |u \cap d_{\xi,m}| < \omega;$$

- (3) if $m \notin \Delta(u)$, then every tree in column m remains Suslin;
- (4) in an active column $m \in \Delta(u)$, every tree $S_{k,1-z(k)}^{\eta,m}$ remains Suslin.

Proof. For the ccc, take an uncountable set of conditions. Pass to an uncountable subset with the same stem s . The branch parts belong to the ccc forcing $\mathbb{B}_{\eta}^*(z)$, so two of them have a common extension. Since the stems are equal, the union of the two side conditions gives a common extension in $\text{Code}_n(z, \eta)$. The size statement follows from CH.

Fix $m \in \Delta(u)$. Once a stem has decided the bit which selects m , the column remains active. The sets which require the column part to meet a given dense subset of $\mathbb{B}_{\eta,m}(z)$ are dense. Hence H_m is generic.

Suppose that $\xi \in Y_{\eta,m,z}$. By the truth lemma, some column condition in H_m forces this statement. The conditions which then put (ξ, m) into the side condition are dense. After this happens, no later 1 is added on $d_{\xi,m}$. Thus $u \cap d_{\xi,m}$ is finite.

Now suppose that $\xi \notin Y_{\eta,m,z}$. Fix $l < \omega$. Below every condition which has activated column m , the set of extensions whose column part decides the statement

$$\xi \in \dot{Y}_{\eta,m,z}$$

is dense. A condition in the generic filter which decides this statement must force nonmembership. Starting with such a condition, choose

$$j > \max\{l, |s_p|\}$$

in $d_{\xi,m}$ and outside the finitely many sets named by F_p . Extend the stem to length $j + 1$, put $s_q(j) = 1$, and put 0 at the other new coordinates. The matrix $\vec{d}$ is pairwise almost disjoint, so this gives a condition. Since l was arbitrary, $u \cap d_{\xi,m}$ is infinite.

It remains to prove the preservation statements. Let T be a tree in an inactive column, or the unselected member of a pair in an active column. To show that T remains Suslin, consider an uncountable set of conditions in $\text{Code}_n(z, \eta) \times T$. Pass to a set with one stem. By independence, the corresponding branch parts together with the T -coordinates contain two compatible members. The union of the two side conditions then gives a common extension. Thus $\text{Code}_n(z, \eta) \times T$ is ccc, and T remains Suslin. $\square$

4.4 The coding relation

For a real u and $m \in \Delta(u)$, let

$$Y_m^u = \{\xi < \omega_1 : |u \cap d_{\xi,m}| < \omega\}.$$

The local version of this set is defined inside a countable model using its local almost disjoint matrix.

Let $\Theta_n(u, z)$ say that every countable transitive model with the required correct M_n -initial segment, and containing u and z , has the following property. For every $m \in \Delta(u)$, the local decoding Y_m^u is a David reshaping code whose odd part decodes a well-founded structure containing:

- (1) one block address η which is independent of m ;
- (2) the column address m ;
- (3) branches through $S_{k,z(k)}^{\eta,m}$ for all $k < \omega$;
- (4) the correct local M_n -initial segment and the statement that the branches record z .

Lemma 4.4. *The relation $\Theta_n(u, z)$ is Π_{n+2}^1 . Hence*

$$\Phi_n(z) \iff \exists u \Theta_n(u, z)$$

is Σ_{n+3}^1 .

Proof. The assertion that a real codes a correct local M_n -initial segment is Π_{n+1}^1 . Once this segment is given, the almost disjoint decoding, David decoding, well-founded collapse, and branch checks are first-order checks in a countable coded model. The definition of Θ_n has one universal real quantifier over such models. It is therefore Π_{n+2}^1 . The last claim follows by existential quantification over u . $\square$

Lemma 4.5 (Global decoding). *If $\Theta_n(u, z)$ holds, then there is a block address $\eta < \omega_1$ such that, for every $m \in \Delta(u)$ and every $k < \omega$, there is a cofinal branch through*

$$S_{k,z(k)}^{\eta,m}.$$

Proof sketch. Fix $m \in \Delta(u)$ and form the global almost disjoint decoding Y_m^u . If the David decoding failed, or if the decoded structure were ill-founded, take a countable elementary hull containing a witness to the failure and collapse it. Condensation gives a countable transitive model with the correct M_n -initial segment. This contradicts $\Theta_n(u, z)$. The same argument for two active columns shows that they decode the same block address. $\square$

Lemma 4.6 (Forcing and persistence). *Let z belong to the current intermediate model and let η be fresh. The forcing $\text{Code}_n(z, \eta)$ adds a real u such that $\Theta_n(u, z)$ holds. This remains true after the later forcing used below.*

Proof. Lemma 4.3 gives the branch pattern and the almost disjoint decoding in every active column. David localization gives the local statement in every countable transitive model. Later coding stages use fresh blocks. The other iterands preserve the trees in the current block. The almost disjoint pattern and all branches already added remain unchanged. Hence $\Theta_n(u, z)$ persists. $\square$

5 The forcing iteration

We now define the final forcing over M_n . Its length is $\omega_1^{M_n}$. We use finite support.

5.1 Bookkeeping

Using the canonical well-order of M_n , fix the least partition

$$\omega_1 = I_{\text{mof}} \cup I_{\text{Coh}} \cup I_{\text{Am}}$$

into three cofinal sets, and fix the least bookkeeping system with the following properties.

- (1) Every nice name for a real, and in particular every nice name for a code in $p(2^\omega)$, is presented at a stage in I_{mof} after the support of the name has appeared. Each such name is presented cofinally often.
- (2) Stages in I_{Coh} add ordinary Cohen reals.
- (3) If a name $\dot{a}$ for a real has appeared, then cofinally many later stages in I_{Am} use the category amoeba forcing computed in

$$L[T_{n+1}, \dot{a}].$$

Since $M_n \models \text{CH}$ and all partial iterations have size at most $\aleph_1$, standard nice-name bookkeeping gives such a system.

We construct

$$\langle \mathbb{P}_\alpha, \dot{\mathcal{Q}}_\alpha : \alpha < \omega_1 \rangle$$

by recursion. Along the construction we also define names

$$\dot{\mathcal{O}}_\alpha$$

for the set of non-atomic measures accepted before stage α .

5.2 The three kinds of stages

Suppose that $\mathbb{P}_\alpha$ and $\dot{\mathcal{O}}_\alpha$ have been defined.

Mof stages. Assume that $\alpha \in I_{\text{mof}}$ and that the bookkeeping presents a $\mathbb{P}_\alpha$ -name $\dot{f}$ for a real. In a $\mathbb{P}_\alpha$ -generic extension, let $f = \dot{f}^{G_\alpha}$.

If $f \notin p_c(2^\omega)$, or if μ_f is not orthogonal to every measure in $\mathcal{O}_\alpha$, let $\dot{\mathbb{Q}}_\alpha$ be the trivial forcing and set

$$\mathcal{O}_{\alpha+1} = \mathcal{O}_\alpha.$$

Otherwise, let η_α be the least ω -block which has not been used at an earlier accepting mof stage. Put

$$z_f = \langle \text{mof}, f \rangle$$

and force with

$$\text{Code}_n(z_f, \eta_\alpha).$$

Let u_α be the self-indexed almost disjoint real added by this forcing. In the extension by the iterand, define

$$g_\alpha = G(f, f \oplus u_\alpha).$$

Then

$$\mu_{g_\alpha} \approx \mu_f \quad \text{and} \quad r_{\text{FT}}(g_\alpha) = f \oplus u_\alpha.$$

Set

$$\mathcal{O}_{\alpha+1} = \mathcal{O}_\alpha \cup \{\mu_{g_\alpha}\}.$$

The tag **mof** is used only at accepting mof stages.

Cohen stages. If $\alpha \in I_{\text{Coh}}$, let

$$\dot{\mathbb{Q}}_\alpha = \text{Add}(\omega, 1).$$

The family $\mathcal{O}_\alpha$ is unchanged.

Category amoeba stages. If $\alpha \in I_{\text{Am}}$, the bookkeeping presents a name $\dot{a}$ for a real. In the intermediate extension, put

$$N_a = L[T_{n+1}, a].$$

Let $\dot{\mathbb{Q}}_\alpha$ be the category amoeba forcing computed in N_a . We use the standard σ -linked presentation which adds a Borel meager set containing every Borel meager set coded in N_a . The family $\mathcal{O}_\alpha$ is unchanged.

At limit stages we take direct limits with finite support. Let

$$\mathbb{P} = \mathbb{P}_{\omega_1}.$$

If $G \subseteq \mathbb{P}$ is generic, put

$$V^* = M_n[G] \quad \text{and} \quad \mathcal{O}_{\text{na}} = \bigcup_{\alpha < \omega_1} \mathcal{O}_\alpha.$$

5.3 Preservation

Lemma 5.1. *For every $\alpha \leq \omega_1$:*

- (1) $\mathbb{P}_\alpha$ is ccc;
- (2) every Suslin tree in a block not used before stage α remains Suslin;

- (3) in a used block, every tree not selected by its coding pattern remains Suslin;
- (4) $|\mathbb{P}_\alpha| \leq \aleph_1$.

Proof. The proof is by induction on α . At a coding stage, Lemma 4.3 gives the ccc and preserves every tree outside the active coding coordinates. Cohen forcing is countable. The category amoeba forcing used here is σ -linked. These forcings preserve every Suslin tree from an unused coordinate. Finite-support iteration preserves the stated properties. The size bound follows from CH in M_n and the fact that every iterand has size at most $\aleph_1$. $\square$

Corollary 5.2. *The forcing $\mathbb{P}$ preserves all cardinals and*

$$V^* \models \text{CH}.$$

Moreover, $\mathbb{P}$ has M_n -cardinality $\omega_1 < \delta_0$. Hence V^ is a small generic extension of M_n , and the canonical M_n -strategy and the comparison facts used below lift to V^* .*

Proof. The forcing is ccc and has size $\aleph_1$. Since $M_n \models \text{CH}$, there are only $\aleph_1$ nice names for reals. Thus $V^* \models 2^{\aleph_0} = \aleph_1$. The final statement follows from $\omega_1 < \delta_0$. $\square$

5.4 Exactness of the reserved code

Lemma 5.3. *In V^* , suppose that*

$$\Theta_n(u, \langle \text{mof}, f \rangle)$$

holds. Then there is a unique accepting mof stage α such that:

- (1) *the stage considered the code f ;*
- (2) *the stage used block η_α to code $\langle \text{mof}, f \rangle$;*
- (3) *$u = u_\alpha$.*

Proof. By Lemma 4.5, the relation $\Theta_n(u, \langle \text{mof}, f \rangle)$ yields a block η such that every column in $\Delta(u)$ contains the branch pattern for $\langle \text{mof}, f \rangle$. An unused block has no branch through any of its trees. In a block used at a mof stage, all active columns record the real presented at that stage, and every inactive column remains Suslin. Hence η is the block used by an accepting mof stage which considered f .

The tag **mof** is used only at accepting mof stages. Once a measure equivalent to μ_f has been added, no later stage can accept f again. Thus the stage α is unique.

At stage α , the active columns are exactly the columns in $\Delta(u_\alpha)$. Every column outside $\Delta(u_\alpha)$ remains Suslin by Lemma 5.1. Since $\Theta_n(u, \langle \text{mof}, f \rangle)$ yields branches in every column in $\Delta(u)$, we have

$$\Delta(u) \subseteq \Delta(u_\alpha).$$

Each of these two sets contains exactly one member of every pair $\{2k+1, 2k+2\}$. Therefore

$$\Delta(u) = \Delta(u_\alpha),$$

and hence $u = u_\alpha$. $\square$

6 The Baire property

We now prove the regularity statement needed for the negative mof argument. Only category forcing is used.

For a real $a \in V^*$, let

$$N_a = L[T_{n+1}, a].$$

The model N_a is the base over which the relative Cohen and category arguments are evaluated.

Lemma 6.1 (Category covering). *For every real $a \in V^*$, there is a Borel meager set $Z_a \in V^*$ such that every Borel meager set coded in N_a is contained in Z_a . Consequently, the set of reals which are not Cohen-generic over N_a is meager.*

Proof. The real a belongs to some intermediate extension because every real name has countable, hence bounded, support. The bookkeeping later places a category amoeba stage over N_a . Let Z_a be the Borel meager set added at that stage. By the definition of category amoeba forcing, every Borel meager set coded in N_a is contained in Z_a . This inclusion is preserved by later forcing.

A real is not Cohen-generic over N_a if and only if it belongs to a Borel meager set coded in N_a . Hence every such real belongs to Z_a . $\square$

Theorem 6.2. *In V^* , every boldface Σ_{n+2}^1 set of reals has the Baire property.*

Proof. Let $A \subseteq \omega^\omega$ be $\Sigma_{n+2}^1(a)$ and put

$$N_a = L[T_{n+1}, a].$$

Work in N_a with Cohen forcing, and let $\dot{c}$ be its canonical name. The Boolean value of the statement “ $\dot{c} \in A$ ” is represented, modulo meager sets, by a Borel set C coded in N_a .

Let z be Cohen-generic over N_a . The forcing theorem gives

$$z \in C \iff N_a[z] \models z \in A.$$

The Martin–Solovay tree T_{n+1} and its absolute complement identify the truth on the right with the truth of $z \in A$ in V^* . The forcing is small relative to the least Woodin cardinal, so the required representation remains absolute; see [MS08, Hof26]. Therefore

$$z \in C \iff z \in A$$

for every Cohen real z over N_a .

It follows that $A \triangle C$ is contained in the set of reals which are not Cohen-generic over N_a . This set is meager by Lemma 6.1. Hence A has the Baire property. $\square$

7 Σ_{n+2}^1 -uniformization in small extensions of M_n

We next derive the uniformization statement from the canonical mouse order. The argument is the relativized Steel analysis used in [Hof26, Proposition 5.12 and Theorem 5.13]; see also [HM26].

Let $N = M_n[H]$ be a forcing extension by a forcing of M_n -cardinality below the least Woodin cardinal, and assume that the canonical strategy lifts to N . Let $s \in \mathbb{R}^N$. Write $M_n(s)$ for the canonical proper class s -mouse with n Woodin cardinals.

Proposition 7.1 (Capture and the good mouse order). *The following statements hold in N .*

(1) *If*

$$N \models \exists u \exists v \theta(s, u, v),$$

where θ is Π_{n+1}^1 , then there are

$$u, v \in \mathbb{R} \cap M_n(s)$$

such that

$$M_n(s) \models \theta(s, u, v).$$

(2) *For $u, v \in \mathbb{R} \cap M_n(s)$ and every Π_{n+1}^1 formula θ ,*

$$M_n(s) \models \theta(s, u, v) \iff N \models \theta(s, u, v).$$

(3) *The canonical mouse order $<_{n,s}$ on $\mathbb{R} \cap M_n(s)$ is a good $\Sigma_{n+2}^1(s)$ well-order. More precisely, if $\theta(s, u, v)$ is Π_{n+1}^1 , then the relation*

$$\text{Least}_\theta^n(s, u)$$

which says that $u \in M_n(s)$ and u is the $<_{n,s}$ -least real for which there is a $v \in M_n(s)$ with

$$M_n(s) \models \theta(s, u, v)$$

is uniformly $\Sigma_{n+2}^1(s)$.

Proof. Codes for the sound initial segments of $M_n(s)$ are described by a $\Pi_{n+1}^1(s)$ condition. The coded premouse must be sound, project to ω , be n -small over s , and satisfy the required Π_n -iterability condition. Steel comparison orders these premice by initial segment and identifies the correctly iterable ones with initial segments of $M_n(s)$.

For the first clause, write a $\Sigma_{n+2}^1(s)$ assertion in the form

$$\exists u \exists v \theta(s, u, v)$$

with θ in Π_{n+1}^1 . The comparison-and-capture argument puts a witness into a countable initial segment of $M_n(s)$. Otherwise, comparison with the canonical construction gives the same contradiction as in the proof of projective correctness for $M_n(s)$.

The second clause is the corresponding correctness statement. Once the witnesses belong to $M_n(s)$, Π_{n+1}^1 truth is computed by the canonical mouse and is preserved to the small generic extension.

For the third clause, assign to a real $u \in M_n(s)$ the least sound initial segment of $M_n(s)$ which contains u , and then use the canonical well-order of that premouse. Comparison makes this a well-order. Since the class of relevant initial segments is $\Pi_{n+1}^1(s)$, its initial segments are uniformly $\Sigma_{n+2}^1(s)$. Least-witness assertions for Π_{n+1}^1 matrices are therefore $\Sigma_{n+2}^1(s)$. $\square$

Theorem 7.2 (Uniformization in small extensions). *Every small generic extension N of M_n as above has boldface Σ_{n+2}^1 -uniformization. In particular, V^* satisfies boldface Σ_{n+2}^1 -uniformization.*

Proof. Work in a small generic extension N of M_n . Let $A \subseteq \mathbb{R}^2$ be a $\Sigma_{n+2}^1(a)$ relation. Choose a Π_{n+1}^1 formula ψ such that

$$A(x, y) \iff \exists z \psi(a, x, y, z).$$

For a real x , put

$$s = a \oplus x.$$

Let $\theta(s, y, z)$ be the formula obtained from $\psi(a, x, y, z)$ after decoding s as $a \oplus x$. Define

$$U(x, y) \iff \text{Least}_\theta^n(a \oplus x, y).$$

By Proposition 7.1(3), the relation U is $\Sigma_{n+2}^1(a)$.

Suppose that $U(x, y)$ holds. Then, for some $z \in M_n(a \oplus x)$,

$$M_n(a \oplus x) \models \psi(a, x, y, z).$$

By Proposition 7.1(2),

$$N \models \psi(a, x, y, z).$$

Thus $A(x, y)$ holds. Hence $U \subseteq A$.

The relation U is single-valued because $<_{n, a \oplus x}$ is a well-order and $U(x, y)$ states that y is the least candidate.

Finally, suppose that the x -section of A is nonempty. Then

$$N \models \exists y \exists z \psi(a, x, y, z).$$

By Proposition 7.1(1), there are witnesses

$$y, z \in M_n(a \oplus x).$$

The set of candidates in $M_n(a \oplus x)$ is therefore nonempty. It has a $<_{n, a \oplus x}$ -least element y_0 , and $U(x, y_0)$ holds. Thus U uniformizes A .

The final model V^* is a small generic extension by Corollary 5.2, so the last statement follows. $\square$

8 The Π_{n+2}^1 maximal orthogonal family

We now verify maximality and definability of the family built during the iteration.

Lemma 8.1. *In V^* , the family $\mathcal{O}_{\text{na}}$ is a maximal orthogonal family of non-atomic measures.*

Proof. Suppose that μ_{g_α} is added at stage α . At that stage, μ_f is orthogonal to every earlier member of the family. Since

$$\mu_{g_\alpha} \approx \mu_f,$$

the new measure is orthogonal to every earlier member. Later accepting stages test orthogonality against the enlarged family. Thus the family is pairwise orthogonal.

For maximality, suppose that $h \in p_c(2^\omega)$ and μ_h is orthogonal to every member of $\mathcal{O}_{\text{na}}$. The real h appears in some intermediate extension. At a later mof stage, the bookkeeping presents a name for h . Since μ_h is orthogonal to the final family, it is orthogonal to the family constructed up to that stage. The stage accepts h and adds a measure equivalent to μ_h . This contradicts the assumption that μ_h is orthogonal to every member of the final family. $\square$

For $g \in \text{dom}(r_{\text{FT}})$, write

$$r_{\text{FT}}(g) = f_g \oplus u_g$$

using the fixed recursive projections.

Theorem 8.2. *The set of codes for the measures in $\mathcal{O}_{\text{na}}$ is lightface Π_{n+2}^1 .*

Proof. Define $g \in \hat{\mathcal{O}}_{\text{na}}$ if and only if:

- (1) $g \in p_c(2^\omega) \cap \text{dom}(r_{\text{FT}})$;
- (2) $f_g \in p_c(2^\omega)$;
- (3) $g = G(f_g, f_g \oplus u_g)$;
- (4) $\Theta_n(u_g, \langle \text{mof}, f_g \rangle)$.

The first three clauses are arithmetical. The last clause is Π_{n+2}^1 by Lemma 4.4. Hence $\hat{\mathcal{O}}_{\text{na}}$ is Π_{n+2}^1 .

If $g = g_\alpha$ was added at an accepting mof stage, then

$$r_{\text{FT}}(g) = f \oplus u_\alpha, \quad g = G(f, f \oplus u_\alpha),$$

and $\Theta_n(u_\alpha, \langle \text{mof}, f \rangle)$ holds. Thus $g \in \hat{\mathcal{O}}_{\text{na}}$.

Conversely, suppose that $g \in \hat{\mathcal{O}}_{\text{na}}$. By Lemma 5.3, there is a unique accepting mof stage which coded $\langle \text{mof}, f_g \rangle$, and its coding real is u_g . The measure code produced at that stage is

$$G(f_g, f_g \oplus u_g) = g.$$

Thus g was added to the family. Therefore

$$\hat{\mathcal{O}}_{\text{na}} = \{g : \mu_g \in \mathcal{O}_{\text{na}}\}.$$

□

Let

$$\mathcal{O} = \mathcal{O}_{\text{na}} \cup \{\delta_x : x \in 2^\omega\}.$$

The set of point measures is Borel. Every point measure is orthogonal to every non-atomic measure, and distinct point measures are orthogonal. A measure which is orthogonal to every point measure has no atoms. Lemma 8.1 therefore implies that $\mathcal{O}$ is maximal in the full space $P(2^\omega)$.

Corollary 8.3. *In V^* , the family $\mathcal{O}$ is a lightface Π_{n+2}^1 maximal orthogonal family in $P(2^\omega)$.*

9 The absence of Σ_{n+2}^1 maximal orthogonal families

We now prove the negative result used in the main theorem.

Proposition 9.1. *Assume that every boldface Σ_{n+2}^1 set of reals has the Baire property and that every total boldface Σ_{n+2}^1 relation on the reals has a boldface Σ_{n+2}^1 uniformization. Then there is no boldface Σ_{n+2}^1 maximal orthogonal family of Borel probability measures on 2^ω .*

Proof. Suppose that $A \subseteq P(2^\omega)$ is a $\Sigma_{n+2}^1(a)$ maximal orthogonal family. Let $x \mapsto \mu^x$ be the Kechris–Sofronidis map for E_I , so that

$$xE_I y \Rightarrow \mu^x \approx \mu^y, \quad \neg xE_I y \Rightarrow \mu^x \perp \mu^y.$$

For $x \in 2^\omega$, put

$$A_x = \{\nu \in A : \nu \not\ll \mu^x\}.$$

Maximality gives $A_x \neq \emptyset$. Moreover, A_x is countable. Indeed, for each $\nu \in A_x$ choose a nonzero probability measure η_ν such that

$$\eta_\nu \ll \nu \quad \text{and} \quad \eta_\nu \ll \mu^x.$$

The measures η_ν , $\nu \in A_x$, are pairwise orthogonal and all lie below μ^x ; hence the ccc of the measure algebra below μ^x implies that A_x is countable.

Define $Q(x, (\nu_i)_{i < \omega})$ by

$$\begin{aligned} Q(x, (\nu_i)_{i < \omega}) \iff & (\forall i < \omega)(\nu_i \in A \wedge \nu_i \not\ll \mu^x) \\ & \wedge (\forall \rho \in P(2^\omega))(\rho \not\ll \mu^x \rightarrow (\exists i < \omega) \nu_i \not\ll \rho). \end{aligned} \quad (\dagger)$$

Every section Q_x is nonempty. To see this, enumerate A_x as $(\nu_i)_{i < \omega}$. If $\rho \not\ll \mu^x$, choose a nonzero probability measure η such that $\eta \ll \rho$ and $\eta \ll \mu^x$. By maximality, some $\nu \in A$ satisfies $\nu \not\ll \eta$. Hence $\nu \not\ll \rho$ and $\nu \not\ll \mu^x$, so $\nu \in A_x$ and $(\dagger)$ holds. Conversely, if $Q(x, (\nu_i))$ and $\nu \in A_x$, apply the second clause with $\rho = \nu$. Since A is orthogonal, $\nu_i \not\ll \nu$ implies $\nu_i = \nu$. Thus

$$Q(x, (\nu_i)) \implies \{\nu_i : i < \omega\} = A_x.$$

The relations $\perp$ and $\ll$ are arithmetical on measure codes. Hence the first clause of $(\dagger)$ is $\Sigma_{n+2}^1(a)$, while the second is Π_1^1 . Therefore Q is $\Sigma_{n+2}^1(a)$. By uniformization, there is a total function

$$F : 2^\omega \longrightarrow P(2^\omega)^\omega$$

with $\Sigma_{n+2}^1(a)$ graph and $Q(x, F(x))$ for every x . For every open $U \subseteq P(2^\omega)^\omega$,

$$F^{-1}(U) = \{x : (\exists \vec{\nu}) [\text{Graph}(F)(x, \vec{\nu}) \wedge \vec{\nu} \in U]\}$$

is $\Sigma_{n+2}^1(a)$, hence has the Baire property. Therefore F is Baire measurable.

If $xE_I y$, then $\mu^x \approx \mu^y$, and consequently

$$\{F(x)(i) : i < \omega\} = A_x = A_y = \{F(y)(i) : i < \omega\}.$$

Define

$$\tilde{F}(x)(\langle i, j \rangle) = F(x)(i).$$

Then $\tilde{F}$ is Baire measurable and

$$xE_I y \implies \tilde{F}(x) E_{S_\infty} \tilde{F}(y),$$

where E_{S_∞} is the coordinate-permutation orbit relation. By Hjorth turbulence, there are a comeager set $D \subseteq 2^\omega$ and a fixed nonempty countable set $C \subseteq P(2^\omega)$ such that

$$A_x = C \quad (x \in D).$$

Since every E_I -class is meager, choose pairwise E_I -inequivalent reals $x_\alpha \in D$, $\alpha < \omega_1$. Fix $\nu \in C$. Then $\nu \not\ll \mu^{x_\alpha}$ for every $\alpha < \omega_1$. Choose nonzero probability measures η_α such that

$$\eta_\alpha \ll \nu \quad \text{and} \quad \eta_\alpha \ll \mu^{x_\alpha}.$$

The measures $(\eta_\alpha)_{\alpha < \omega_1}$ are pairwise orthogonal and all lie below ν , contradicting the ccc of the measure algebra below ν . Hence no such A exists. $\square$

10 Proof of the main theorem

Proof of Theorem 1.1. Let $V^* = M_n[G]$ be the extension by the iteration from Section 5. Corollary 5.2 gives cardinal preservation and CH. Theorem 6.2 gives the Baire property for all boldface Σ_{n+2}^1 sets. Theorem 7.2 gives boldface Σ_{n+2}^1 -uniformization. Corollary 8.3 gives a lightface Π_{n+2}^1 mof. Proposition 9.1, applied to the preceding two conclusions, rules out every boldface Σ_{n+2}^1 mof. $\square$

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